SHASHANK CHIKATTIMALA

Senior Software Engineer | AI / LLMOps Engineer & Architect
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PROFESSIONAL SUMMARY

Senior engineering leader who has shipped enterprise systems for Lowe's, Walmart, Bridgewater and the State of Arkansas. Currently lead a front-end team and own architecture, release quality and the Docker/Kubernetes/CI-CD plumbing underneath.

Focused on production AI: Enterprise Agentic RAG, LLM guardrails with measured evaluation, and hardened LLM APIs that keep features from breaking in production.

Care about latency, cost per token, and systems that fail loudly. Mentor engineers, set standards, make the build-vs-buy calls.

TECHNICAL SKILLS

• AI Orchestration & Frameworks: LangGraph, LangChain, Enterprise Agentic RAG, agent routing, hybrid search (BM25 + dense, RRF), reranking (FlashRank), LLM gateway (Portkey)

• LLMs & Foundations: gpt-oss-120b and gpt-oss-20b (via Groq), Claude Haiku 4.5, GPT-4o-mini, all-mpnet-base-v2, all-MiniLM-L6-v2 and text-embedding-3-small embeddings, Hugging Face

• AI Safety, Guardrails & Evaluation: NeMo Guardrails, LangSmith, Ragas, retrieval evaluation (hit@k, MRR, bootstrap confidence intervals), CI regression checks

• MLOps / LLMOps / Compute: Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Azure DevOps, Octopus Deploy, Ansible, Gradle, Bash, Hybrid Cloud / On-Prem

• Data & Vector Databases: Qdrant, Chroma, pgvector, SQL, DB2, Cosmos DB, MongoDB

• Core Languages & Backend: Python, TypeScript, JavaScript, Java, XML, FastAPI, Node.js, Express, Swagger, Apollo / URQL, REST, GraphQL

• Cloud & Frontend: AWS (Lambda, SQS, SNS, Kinesis, S3, EC2, RDS, DynamoDB, IAM, CloudWatch, CloudFormation, CloudFront, Route 53), Azure (AKS, Functions, Data Factory, Blob Storage, VMs), React, Angular / AngularJS, Next.js, SASS, Tailwind, Material UI, Storybook, Webpack, Babel, PWA, Redux, Redux Toolkit, Zustand, Module Federation

• Testing & Tooling: Jest, Enzyme, React Testing Library, Mocha / Sinon, Cypress, K6, Sauce Labs, Git, Bitbucket, Jira, Confluence, Splunk, Postman, Agile / Scrum, pytest

• Leadership & Soft Skills: Technical leadership, mentoring, cross-functional communication, stakeholder alignment, code review, problem solving, architecture decisions

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Lowe's, Charlotte, NC | 06/2023 - Present

• **Team and delivery:** Lead a front-end team building the pre- and post-fulfillment order management platform (React, Next.js). Communicate daily with product managers, UX designers and upstream/downstream teams to hold delivery dates and resolve blockers. User engagement up 40%.

• **Monolith to micro frontends:** One monolith meant every team shipped on the same schedule. Moved it to Webpack Module Federation with runtime loading and a versioning strategy that keeps rollouts backward compatible. Teams now ship independently.

• **Page load:** Next.js ISR, static optimization and dynamic imports gave 25% faster loads. Code-splitting with Suspense took another 20% off initial load.

• **Quality gates:** Set up Jest, React Testing Library, Cypress and K6 performance thresholds. Production bugs down 20%. Automated linting and deploys cut the release cycle 30%.

• **State and process:** Adopted Redux for complex state and built reusable components with hooks and Context API. Joined CAB meetings, sprint planning and retrospectives, flagged technical debt, and ran stage-wise releases including in-store dark-mode testing.

• **Leadership and mentoring:** Set front-end standards, ran code reviews, pair sessions and weekly learning sessions, mentored junior engineers, and contributed to the shared Lowe's UI component library. Standardized Docker environments and Jenkins pipelines with DevOps.

Sr. Full Stack Developer | Walmart (via EPAM Systems), Bentonville, AR | 11/2020 – 05/2023

• **Release engineering:** Build and release processes differed across divisions. Standardized them on Azure DevOps, Octopus Deploy, Ansible, PowerShell, Bash and Gradle, with scripted DEV/QA/UAT/PROD deployments.

• **Kubernetes and IaC:** Built AKS and self-hosted Kubernetes clusters with Terraform and Ansible. Dockerized test environments. Designed Azure Data Factory pipelines for data ingestion and publishing, and managed VMs and Blob Storage.

• **Applications:** Built a role-based React app (Context API, Redux) and Node/Express services. Tested with Jest, Mocha, Sauce Labs. Debugged production with Splunk.

Sr. Front End Developer | Bridgewater Associates (via EPAM Systems), Norwalk, CT | 04/2020 - 10/2020

• **Search for an investment firm:** Built a tokenized search (React, TypeScript, CodeMirror) with fuzzy matching and a debounced hook. IndexedDB caching skipped repeat database calls.

• **Quality and delivery:** Ag-Grid and Highcharts for data display, BackstopJS visual regression, CI/CD on AWS (EC2, S3, RDS, DynamoDB).

Full Stack Developer | Walmart (via eSystems Inc), Bentonville, AR | 10/2019 - 03/2020

• Built the associate preboarding app (React, Material-UI, Node, Azure Functions, Cosmos DB) so new hires submit details before day one.

Full Stack Developer | State of Arkansas DHS (via eSystems Inc), Little Rock, AR | 06/2017 - 10/2019

• **Configurable intake:** Built a drag-and-drop form builder in React (MongoDB templates, MySQL referrals) so power users change forms without a deploy.

• **Serverless and offline:** Node APIs on AWS Lambda with Cognito, SNS, SQS and Kinesis. PWA with service workers, redux-persist, web push notifications and IndexedDB works on dead networks.

• **Curam:** Customized IBM Curam (Java, XML, DB2, WebSphere) for healthcare eligibility, including the DHS S.A.V.E. immigration check. Built React/Redux portals and a React Native proof of concept.

• **Incident and accident reporting:** Built a React/Redux incident report system with reusable form components and background file uploads, on Node/Express REST services with MongoDB (Mongoose).

Full Stack Developer | Peta Digit Inc., India | 05/2011 - 07/2015

• Built an e-commerce site, a role-based CMS and a feedback dashboard (AngularJS, Node.js, jQuery / jQuery UI, JSP, SASS), working directly with business users and UX.

AI PROJECTS (GITHUB)

• **kube-rag: Agentic RAG with measured retrieval** (Python, LangGraph, FastAPI, Qdrant, FlashRank, NeMo Guardrails, Portkey, pytest, GitHub Actions): Built a 65-question evaluation set with quote-verified labels, LLM-free hit@5 and MRR metrics, and paired bootstrap confidence intervals. Showed the cross-encoder reranker adds +0.077 hit@5 and +0.106 MRR (95% CIs exclude zero), and that the planner's query rewrite cost 0.09 MRR until searching with the user's own words fixed it. Diagnosed a daily rate limit that had silently failed 28 of 75 evaluation requests. 92 tests with CI.

• **guardrail-eval-ci: Measured guardrail evaluation in CI** (Python, FastEmbed, GitHub Actions, pytest): Reimplemented NeMo Guardrails' embedding gate and checked it against NeMo's own similarity scores

(largest difference 1.4×10^{-9}). Built a 340-message labelled set, a threshold rule written before the sweep, and CI regression checks that never score the test set. On the test set the gate blocks 97% of attacks (29/30, 95% CI 83–99%) with no legitimate questions wrongly blocked; off-topic recall is 23%, reported as a limit.

• **production-rag-1: Hardened LLM API** (FastAPI, LangGraph, slowapi, Docker): Rate limiting, prompt-injection and PII checks, TTL cache, fallback model and output validation. Wrote 56 tests that found and fixed a crash in the retry graph, error text returned and cached as an answer, and 500 responses that leaked internals.

• **RAG engineering lab** (LangChain, Chroma, pgvector, BM25): Runnable demos of semantic chunking, cost controls and tracing. Built hybrid BM25 + dense search with 18 offline tests, and measured it against dense-only search on 65 labelled questions: no reliable gain (MRR +0.037, 95% CI " 0.034 to +0.110).

EDUCATION & CERTIFICATIONS

• M.S., Computer Science, University of Michigan-Flint, 2017

• B.Tech, Information Technology, K.S. Raju Institute of Technology and Sciences, JNTU, 2011

• IBM Curam Certified Application Developer v7, 2018